Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

A.A. 2026/2027

Lecture 10 – Multimedia Analysis

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QUALITY ENHANCEMENT



Image/video enhancement

ENHANCING IMAGES/VIDEOS

- Images/videos are usually taken in adversarial conditions (e.g., video surveillance at night)
- Need to process evidences so that quality is enhanced.
- Software or source code can be used

Softwares:

Adobe Photoshop
Photoshop remains the industry standard for professional photo editing.
Adobe Lightroom
Lightroom is ideal for photographers who manage large volumes of images.
Capture One: Praised for its excellent color management and RAW processing capabilities.
Corel PaintShop Pro: A strong alternative to Photoshop, offering powerful editing tools with a one-time purchase.
Affinity Photo: A cost-effective, single-purchase software with professional-level editing features.

Free and Open-Source:

GIMP (GNU Image Manipulation Program)
Darktable and RawTherapee
These applications are designed for processing RAW files.

DATA ENHANCEMENT – IMAGE/VIDEO

Tools for image/video enhancement in multimedia forensics




- *Sharpening* – Makes edges more clear and distinct.
- *Video stabilization* – Reduces the amount of movement in the video
- *Distortion correction*
- *Perspective distortion correction*
- *Deblurring*
- *Noise reduction*




- *Masking* – Covers the face or areas of the video that may protect a witness, victim or law enforcement officer.
- *Enhancing* contrast, reducing blurring equalizing color
- *Magnifying* resolution of image

Related to older video technology systems

- *Interlacing* –interlaced scanning is used in analog system to record images. A process called de-interlacing may be used to retrieve the information in both fields of video.

- *Demultiplexing* – Allows for isolation of each camera. In CCTV systems, combi multiple video signals into a single signal or separate a combined signal.



CONTRAST ENHANCEMENT

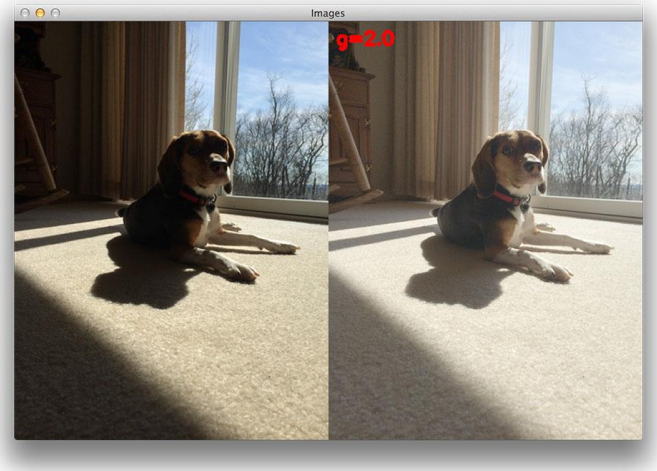


Image pixels are transformed

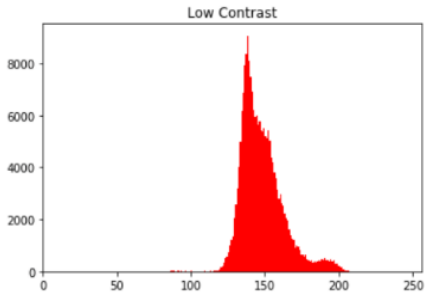
$$I^g(x,y) = (I(x,y))^g$$

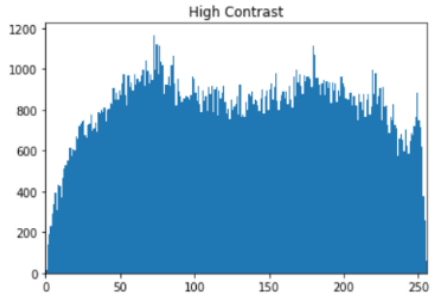
$$I'(x,y) = (I^g(x,y) - m)/(M - m)$$

Given parameters g,m,M, we can convert one image into another.









AMPED SOFTWARE

CONTRAST ENHANCEMENT

Histogram equalization

```
r_image, g_image, b_image = cv2.split(image_src)
r_image_eq = cv2.equalizeHist(r_image)
g_image_eq = cv2.equalizeHist(g_image)
b_image_eq = cv2.equalizeHist(b_image)
image_eq = cv2.merge((r_image_eq, g_image_eq, b_image_eq))
```

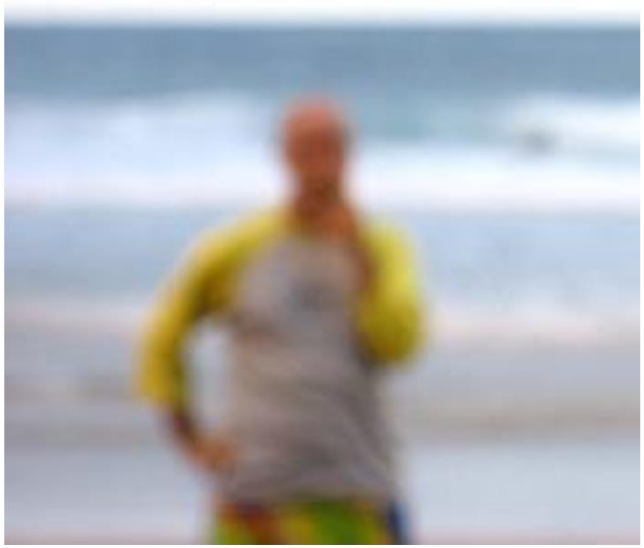
Contrast enhancement

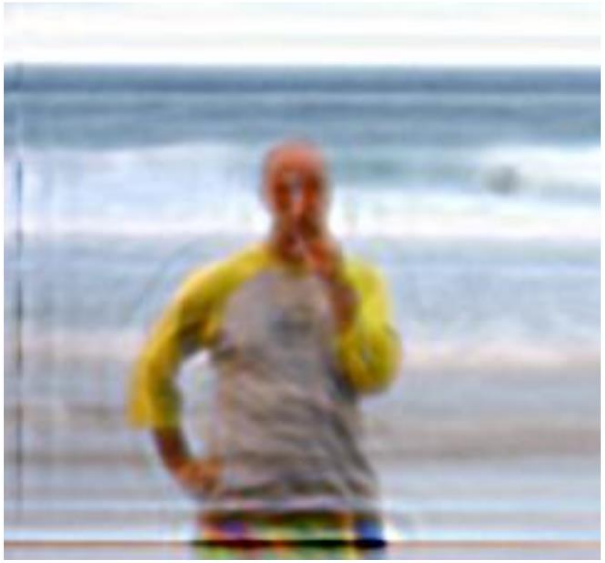
```
image_flattened = image_matrix.flatten()
image_hist = np.zeros(bins)
# histogram creation - you can use cv2.calcHist
for pix in image_matrix:
        image_hist[pix] += 1

cum_sum = np.cumsum(image_hist) #cumulatve sum
norm = (cum_sum - cum_sum.min()) * 255
n_ = cum_sum.max() - cum_sum.min() #see equations
uniform_norm = norm / n_
uniform_norm = uniform_norm.astype('int')
```

You could use CLAHE (you do not have full invertibility)

SHARPENING

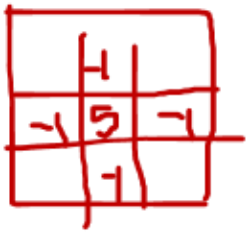




Sharpening can be performed by applying a filtering on the image

```
kernel = np.array([[0, -1, 0], [-1, 5,-1], [0, -1, 0]])
image_sharp = cv2.filter2D(src=image, ddepth=-1, kernel=kernel)
```

i.e. use a highpass filter



AMPED SOFTWARE

MAGNIFICATION

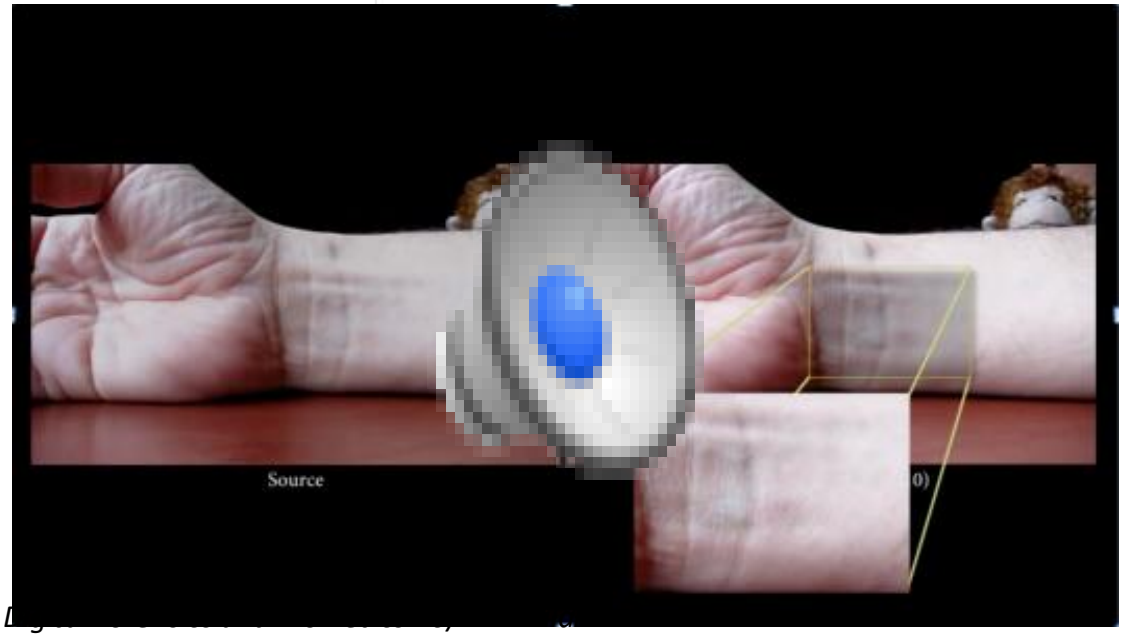




It is possible to increase the size and the level of clarity of an image

Y. Romano, J. Isidoro, P. Milanfar, "**RAISR: Rapid and Accurate Image Super Resolution**", arXiv:1606.01299

When video is involved, it is possible to use multiple frames and do it on motion.



Hao-Yu Wu, Michael Rubinstein, Eugene Shih, John Guttag, Frédo Durand, William T. Freeman
Eulerian Video Magnification for Revealing Subtle Changes in the World
ACM Transactions on Graphics, Volume 31, Number 4 (Proc. SIGGRAPH), 2012

MAGNIFICATION

Eulerian Video Magniification

```
pip install PyEVM
```

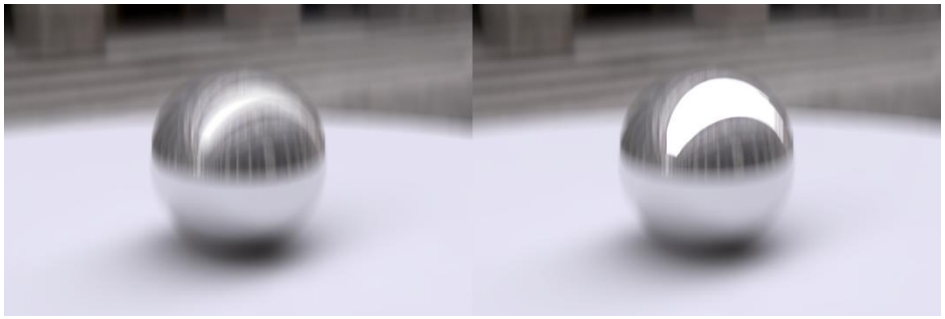
Or for single image you use magnify

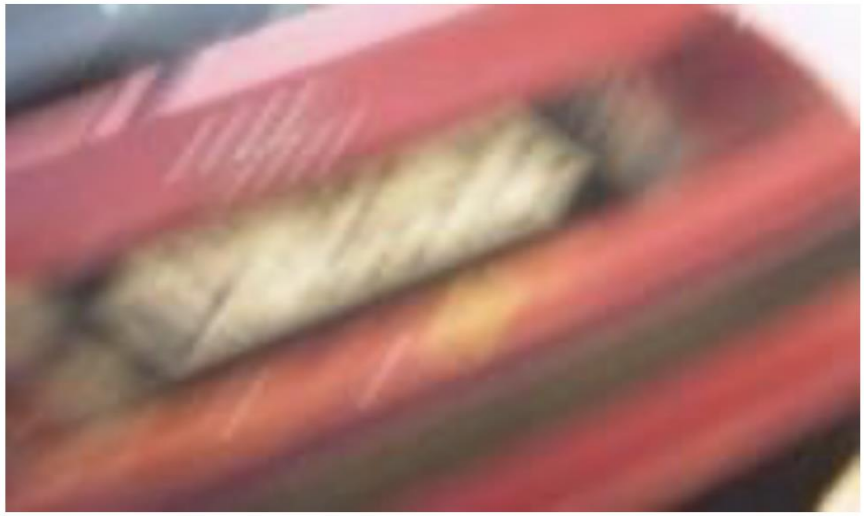
```
from wand.image import Image
Image.magnify()
```

Or rescale

```
# Scaling up
scaleUp = cv2.resize(source, None, fx= scaleX,fy= scaleY,
                interpolation=cv2.INTER_LINEAR)
```

BLURRING COMPENSATION







AMPED SOFTWARE

PERSPECTIVE CORRECTION

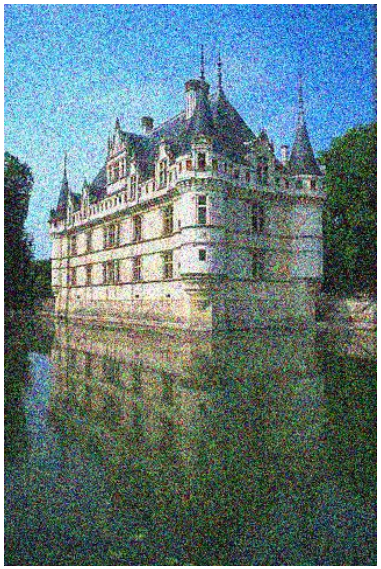




```
matrix = cv2.getPerspectiveTransform(pts1, pts2)
result = cv2.warpPerspective(frame, matrix, (500, 600))
```

AMPED SOFTWARE

NOISE REDUCTION





Classical techniques: adaptive filtering

Neural network strategies

K. Zhang, W. Zuo, Y. Chen, D. Meng and L. Zhang, "Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising," in *IEEE TIP 2017*





LENS DISTORTION CORRECTION

Many cameras use highly-spherical non-spherical geometry lenses: strong distortion due to curvatures and non-idealities

Strong distortion to be corrected









FRAME INTEGRATION





AMPED SOFTWARE

Audio Enhancement



AUDIO ENHANCEMENT

Audio enhancement refers to the process of improving the quality, clarity, and intelligibility of audio signals. Denoising, a crucial aspect of audio processing, is essential for removing unwanted noise, distortions, or artifacts from audio recordings.

Noise to be removed/isolated:

- Background (persistent, low level due to environment, air coolinfg, fans, …)
- Impulse noise (clicks, cracks due to electric interference, mic handling, …)
- Electrical noise (hum, buzzes due to electrical devices and recoders, mics, …)
- Reverberation (due to reflections, rooms, ….)

.

FORENSIC AUDIO ENHANCEMENT AND ANALYSIS

Tools for audio enhancement in multimedia forensics

- *Noise reduction*
- *Speaker identification*
- *Speech identification*
- *Environment estimation/localization*
- *Event reconstruction*
- *Authentication*

Examples:

Watergate investigation, 18½-minute gap in an audio recording of President Richard Nixon discussing the Watergate break in with his Chief of Staff. Analysis determined which recorder made the erasure and how many different erasures were made. The level of AC hum recorded to tape even provided details about the acquisition location.

AUDIO ENHANCEMENT



Examples: *Watergate investigation*

Very frequently, audio is cleaned and amplified and processed by human listening: human brain performs a filtering and focusing (be aware of pareidolia)

Automatic detection can be used for monitor and surveillance (count how many vehicles, …)

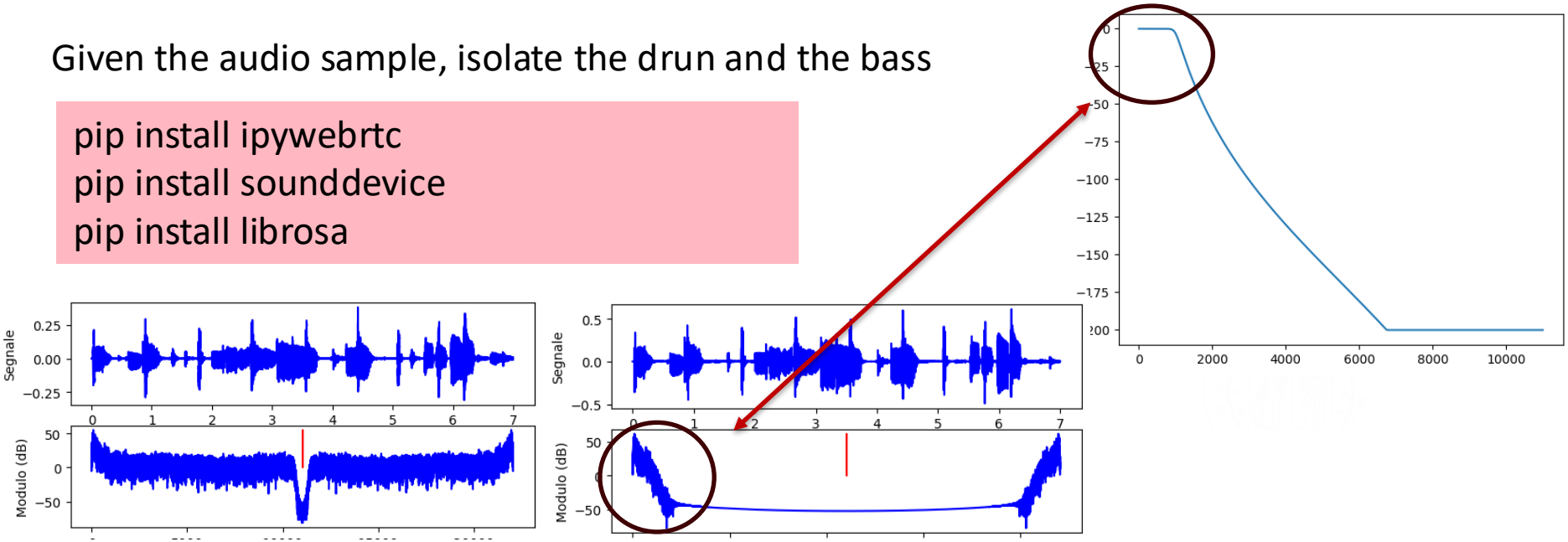
Tasks:

- Critical listening
- Equalization: select the frequency you want
- Dynamics processing: expand or siften the components
- Reverb and de-reverb
- Remove or reduce unwanted sounds (air conditioning, electric equipment, …); not continuous sounds are more troublesome (e.g., dog barking) -> *spectrogram analysis*: sounds can be localized and removed
- Electronic measurements and visualization

EXAMPLE: FREQ SELECTION

Given the audio sample, isolate the drun and the bass

```
pip install ipywebrtc
pip install sounddevice
pip install librosa
```



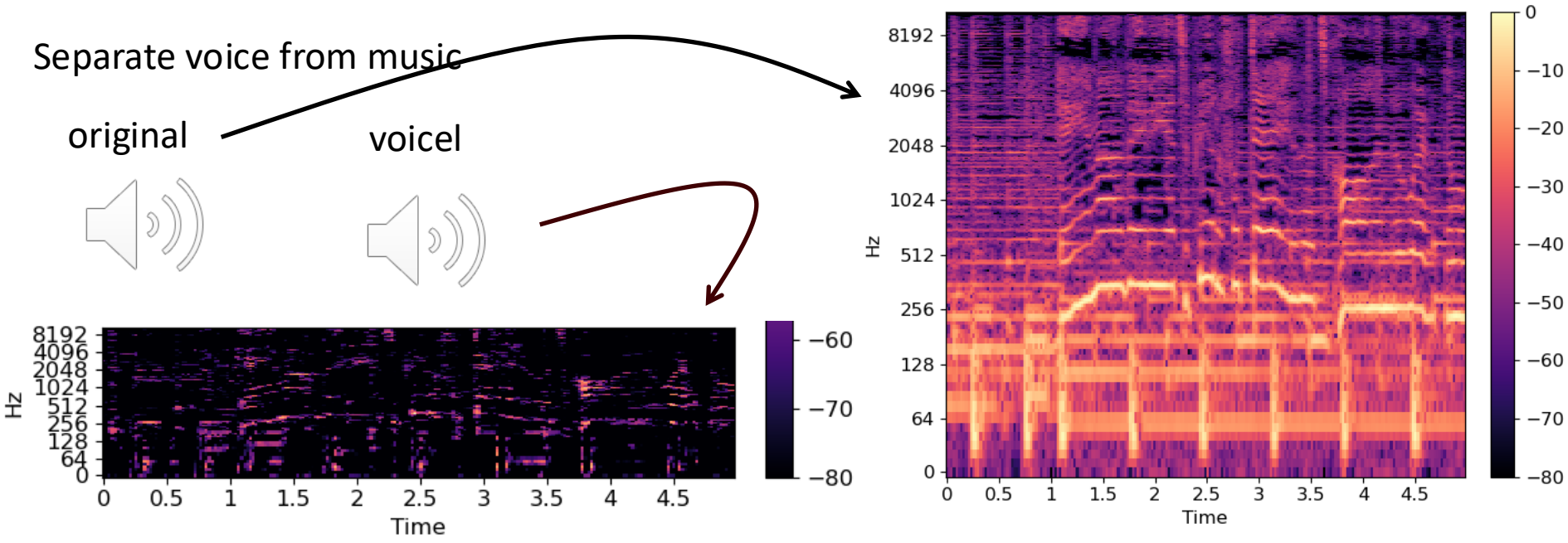
original

Filtered (bass)

Filtered (drum)

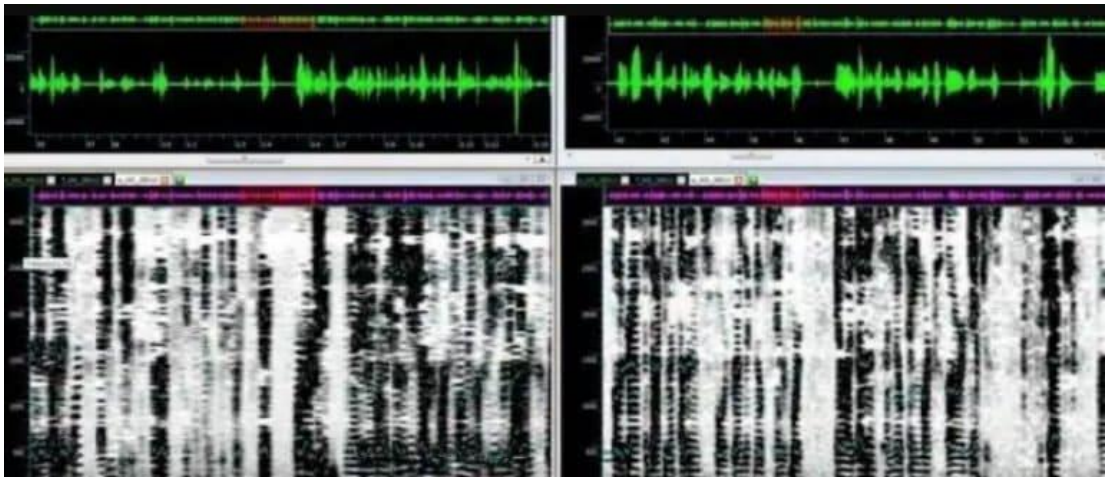
```
#create filter with 10 coeffs with cut freq  1000 Hz
sos = sig.butter(10, 1000, fs=Fs, btype='lowpass', analog=False, output='sos')
w,H = sig.sosfreqz(sos, fs=Fs)
plt.plot(w, 20*np.log10(np.maximum(1e-10, np.abs(H))))
y = sig.sosfilt(sos,x)*2.0
```

EXAMPLE: VOCAL SEPARATION




```
S_full, phase = librosa.magphase(librosa.stft(x))
S_filter = librosa.decompose.nn_filter(S_full,
                       aggregate=np.median,metric='cosine',
                            width=int(librosa.time_to_frames(2, sr=Fs)))
S_filter = np.minimum(S_full, S_filter)
margin_i, margin_v = 2, 10
power = 2
mask_i = librosa.util.softmask(S_filter,margin_i * (S_full - S_filter),power=power)
mask_v = librosa.util.softmask(S_full - S_filter,margin_v * S_filter,power=power)
S_foreground = mask_v * S_full
S_background = mask_i * S_full
```

SPEECH COMPARISON



- compares a known voice sample to an unknown voice sample.
- likelihood of whether the speakers are the same or different.
- Also used to transcript messages
- Enhancement can be used to highlight whispering and distant voices

EXAMPLES AND MORE REFERENCES

- Example of audio enhancement:

	First.	After
Handheld mic in a pocket *AC noise, wall reflections*		
Wireless mic to squad video corder *traffic, poor quality*		
911 call *Handling noise*		
Miniature video camera Distortion reduction		

- Audio forensics overview link

EXAMPLE: ANALYZING AN IMAGE

EXAMPLE

Enhance the visibility of the face of the person in this frame



EXAMPLE



Hypothesis:

- Taken during night or evening hours (low contrast and range)
- Presence of noise
- Blurring is affecting the image
- Resolution is small

Possible solution:
-denoise the image via `fastNlMeansDenoisingColored` from OpenCV2
-perform some gamma correction
-perform some deblurring with a sharpening filter
-select the detail of face
-enlarge the detail by 3 times



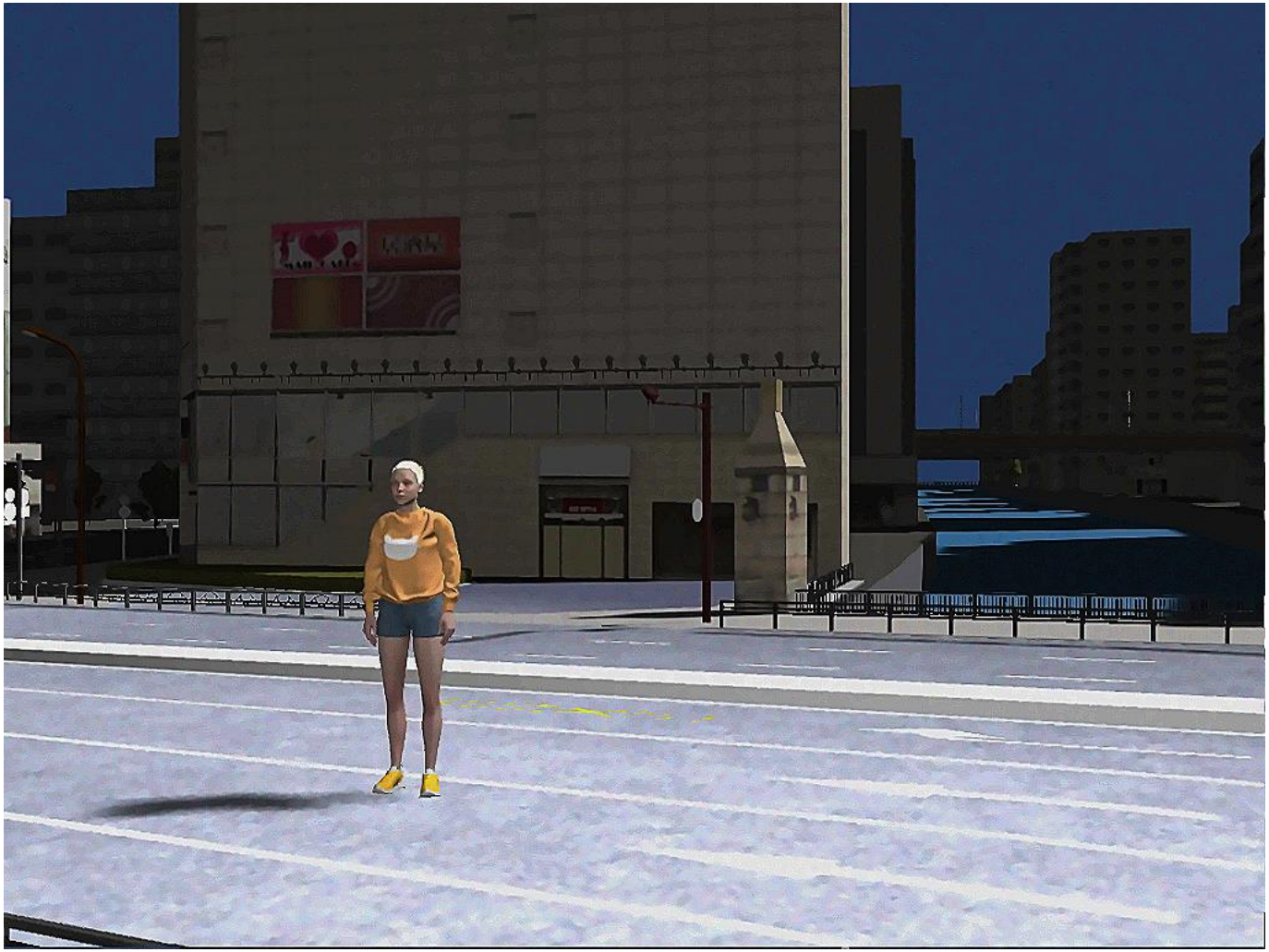
DENOISE



GAMMA CORRECTION



SHARPENING



DETAIL



ENLARGE



CODE

```
Idenoised = cv2.fastNlMeansDenoisingColored(I,None,4,4,3,7)

image_eq = (Idenoised - Idenoised.min())/(Idenoised.max()-
Idenoised.min())*255
image_eq = image_eq.astype('uint8')

#gamma via lookup table
invGamma = 1.1
table = np.array([((i / 255.0) ** invGamma) * 255
                  for i in np.arange(0, 256)]).astype("uint8")
image_eq = cv2.LUT(image_eq, table)

kernel = np.array([[0, -1, 0], [-1, 5,-1], [0, -1, 0]])
image_sharp = cv2.filter2D(src=image_eq, ddepth=-1, kernel=kernel)

image_detail = image_sharp[350:420,290:360,:]

width = image_detail.shape[1]*3;
height = image_detail.shape[0]*3;
dim=(width,height)

Ifinal = cv2.resize(image_detail, dim, interpolation=cv2. INTER_LANCZOS4)
```

How to write a report

How to write ...

The report is usually divided into the following parts

- Synthesis
  - Objective
  - Summary of conclusions

- Relation about the evidences analyzed
  - What did you get?
  - What did you use (softwares, tools, ..)?
  - What do you hand in?

  -list of appendixes
  -ISO/IEC 27037 declaration (reproducibility)



- How do you extract the evidence you analized

- Analysis

- Conclusions

- Your CV

SYNTHESIS: OBJECTIVE, ACTIVITIES AND CONCLUSIONS IN SHORT

What are the questions that are posed/answered by the relation?

- Write clearly what your analysis is intended for
- Better to have a style like "hypothesis testing"
- Define two or three synthetic question so that the conclusions are clear.

What have you done?

- Write what activities you carried on
- Better a short bu clear list

What can you conclude about the evidence ?

- Answer the questions you posed at the beginning
- Be synthetic but clear and detailed.



EXAMPLE

Objective: This relation is intended to improve the quality of an image under analysis. In particular, we want to verify if it is posible to

- Improve the quality of the image reducing noise and blurring
- Increase the detectability of the face of the person depicted in the frame.



Summary of the activities:
As regards the objectives of this consulting, the activity has been carried on as follows:

- Denoising and quality improvements of the picure
- Face extraction
- Face magnification

Conclusions: The operations that we have carried on on the frame has allowed to improve the quality of the image. More precisely, we can answer that:

- Image quality can significantly improved
- We can detect a face of a caucasian woman, short haired, whose identity could correspond …

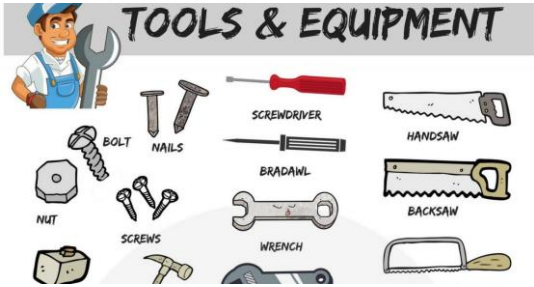
RELATION

What did you get? **Received material**

- Names, characteristics and details about the original files that you have analyzed/processed (extension, format, frame rate, ….)
- Names of softwares that you have received (e.g. some playing tools in case video sequence has some proprietary format)
- Better if you report also hash signatures for verification

What did you use? **Adopted softwares**

- Name all the tools that you have used in the analysis
- Better if you use open source solutions
- Name also OS and hardware tools (PCs) you have used



What do you hand in? **Produced material**

- If you produce a folder with videos, software tools, the report, etc… document all these stuff
- Report also a file with a hash signature for all the files.
- Include also the copy of the relation

EXAMPLE

Received material

File: `image_to_be_analyzed.png`

```
Format: PNG
Resolution: 1024x768
MD5:
SHA-1:
```

Adopted softwares

Within the analysis we have done, the following softwares and tools were adopted.

- Laptop MacBook Pro (13-inch, 2020) with operating system macOS Monterey 12.5.1 (21G83)
- Tools *md5* e *shasum* on macOS to compute hash MD5 e SHA-1.
- Python version 3.8.1 available at https://www.python.org/downloads/release/python-381/
- Library opencv-python version 4.4.0 disavailable at https://opencv.org/opencv-4-4-0/
- Library matplotlib available at https://matplotlib.org/.
- Library shutil available at https://docs.python.org/3/library/shutil.html
- Library numpy available at https://numpy.org/
- Library mpldatacursor available at https://pypi.org/project/mpldatacursor/
- Script *testo.py,* created to analyze the image

Handed-in material (describe the folders and subfolders with their content)

- /softwares: …
- /original_image
- /created image
- sha files, md5-files

EXAMPLE

Appendixes

The relation is made of N pages, it is linked to a digital archive and includes a final appendix

Reproducibility

The analysis follows the standard for handling digital evidences ISO/IEC 27037 and all the results can be replicated using the included software.

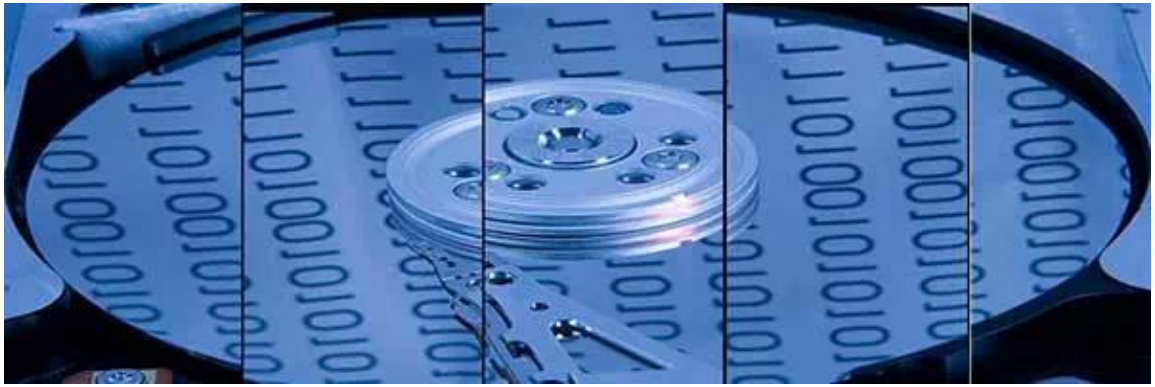
How do you extracted the evidence?

Acquisition process

Report the way you extracted the material (i.e. frames in a video sequence)

If you select some parts or instants, explain why (e.g. *the subject of interest entered the field of view of the camera at …*)

Explain why you used some specific tools and their configuration



ANALYSIS

Write down what you have done ….

Report all the parameter values and details of your analysis

Report intermediate results

Be detailed but clear and simple.



EXAMPLE

Analysis

Performed a denoising using

```
Idenoised = cv2.fastNlMeansDenoisingColored(I,None,4,4,3,7)
```

Parameter were selected ….

This led to the result ….

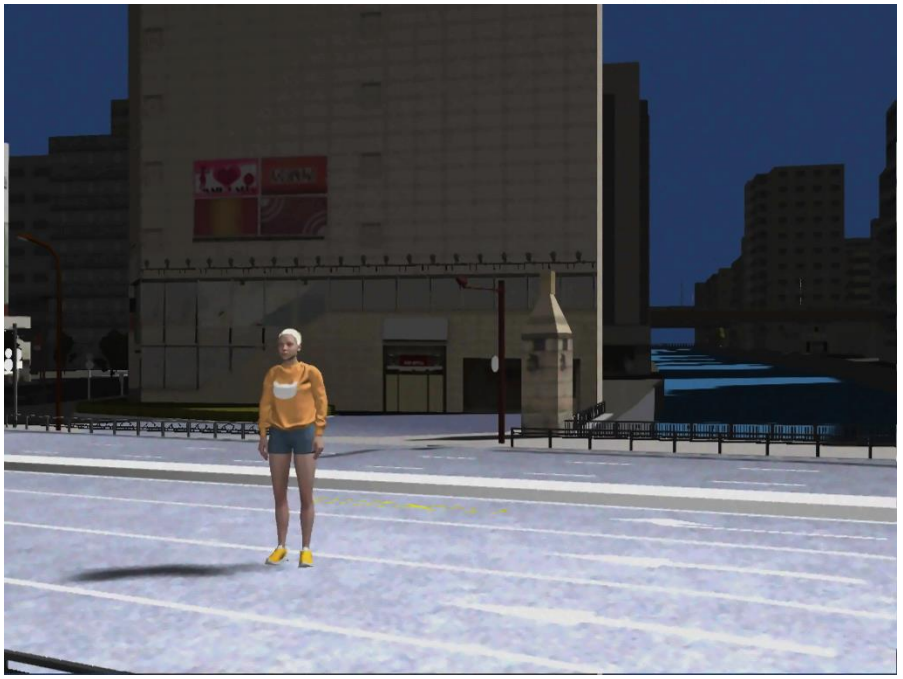


EXAMPLE

Analysis (cont'd)

Then, gamma correction is performed with min=0.1 max=0.5 gamma=1.1 using the operations ….

Parameter were selected …..

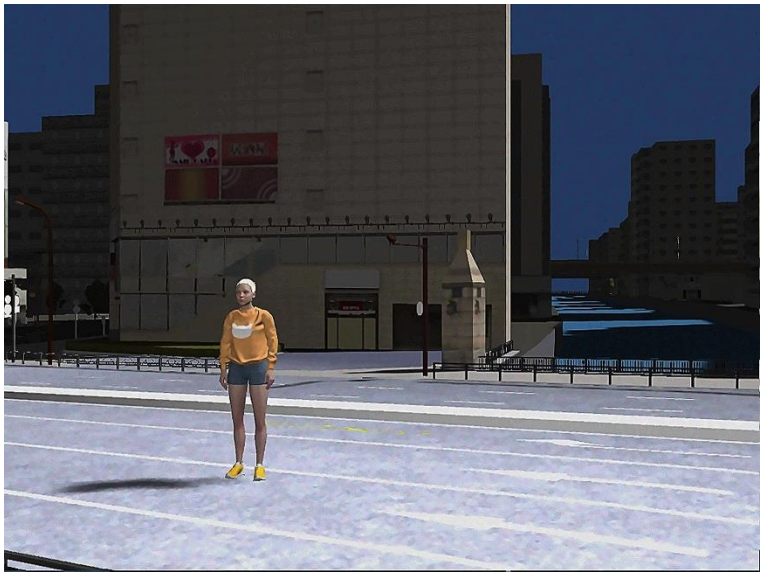


EXAMPLE

Analysis (cont'd)

Then, sharpening was applied with ….

Parameter were selected …..



A detail within the window sized 70x70 at coord (350,290) (first coord is vertical) is eected since ….

Detail is magnified 3 times with command ….





Conclusions

State more clearly and finally what you have anticipated

Signature and date

Example

We have identified the subject as …

It is possible to state that ….

Padova 29/05/2023

Name and Surname



TO AVOID MISTAKES …

- **Avoid personal opinions**

  Objectivity is key in forensic analysis. Stick to the facts, also in the writing style. Drop phrases like "I believe" or "In my opinion.» ….

- **Motivate all claims**

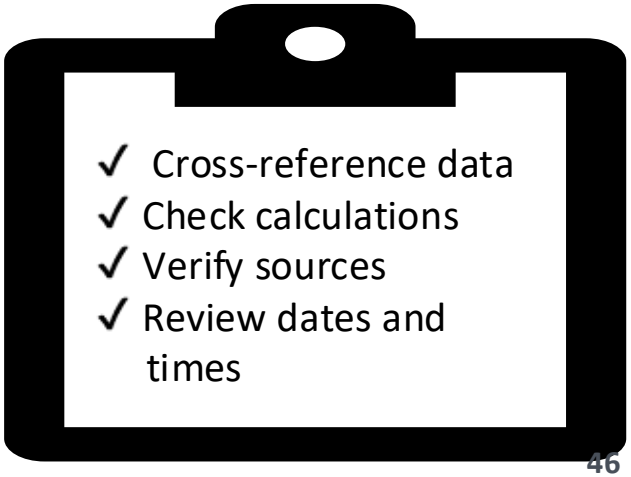
  Describe your your methods clearly, explain how evidence supports your conclusions, use data, charts, or images as backup
  e.g. Amanda Knox case

- **Report must be consistent**

  Otherwise doubts may arise (e.g. no mixed units, different date formats, change of terms, …)
  e.g. OJ Simpson case

- **Check data**

  -expert review
  -double check data
  -track changes and version
  -be careful to legally-relevant details



LEGAL ISSUES

- **Validity and authentication**
  - clear chain of custody
  - come from a qualified expert
  - use scientifically valid methods
  - evidence was not tampered

What	Who	When	Where
Collection	Officer John Brown	2023-06-15, 10:30 AM	Crime scene
Storage	Evidence clerk	2023-06-15, 2:00 PM	Locker #42
Analysis	Lab tech Sarah Lee	2023-06-16, 9:00 AM	Forensic lab

- **Privacy issues**

  often contain sensitive info. How do you balance openness with privacy?
  - use initials or case numbers
  - cut out personal info that's not case-related
  - follow data protection laws for digital evidence